



Eclipse Science: The EZIE Way

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Background

The earth is constantly being bombarded by radiation from the sun. The earth’s magnetic field, or *magnetosphere*, protects us from these particles by deflecting them away. Interactions between the sun and the earth’s magnetic field result in interesting phenomena that are not fully understood. For instance, the aurora borealis, which are observed on earth and other planetary bodies, occur when solar particles are accelerated at the earth’s poles and interact with molecules in our atmosphere! Moreover, the sun’s magnetic field is strong enough compared to earth’s that it can compress its magnetic field lines on the day side of our planet. Other interesting interactions are beyond the scope of this poster, but supplemental materials can be located using the QR code.

Our sun is approximately 400 times larger than our moon, and 400 times as far away, which allows the moon to completely eclipse the sun when there is proper alignment. This offers an interesting opportunity to observe electromagnetic phenomena. **We sought to ask: how is the Earth’s magnetic field impacted during a total solar eclipse?**

April 8, 2024, Eclipse Overview:
Duration: 13:59 -16:29 EDT
Totality: 15:13 - 15:17 EDT

Methods

An EZIE-Mag kit (Fig. 2) with an RM3100 magnetometer (same model used on NASA’s Artemis mission!) and MCP9808 temperature sensor were used to collect magnetic field and temperature readings during a solar eclipse. The equipment was placed on camping chair far away from electromagnetic interference and set to run for eclipse’s duration.

The magnetometer uncertainty was calculated by finding the standard error of the mean (SEM) for magnetic field components using the number of samples and their standard deviations during an earlier calibration period. The final uncertainty in the overall mean magnetic field strength is determined by propagating the errors from individual SEMs, accounting for their respective contributions to the total magnetic field measurement. Our dominant source of uncertainty was a substantial gap in data and abnormal fluctuations recorded right before the eclipse during bootup. Scan the QR code for more info.

Results

Unfortunately, due to a power supply issue, data was not recorded in the early parts of the partial eclipse, which can be seen in the graphs. Nonetheless, we measured a substantial change in magnetic field strength and temperature as a function of time, quantitatively described in Table 1.

The graphs depict steady downward trends of temperature and magnetic field as the sun became increasingly eclipsed by the moon. Figure 3 shows temperature vs time readouts from another EZIE-Mag at a nearby location along the path of totality.

* Calibration period high (24.56 °C) minus totality low (16.625 °C). See Discussion for explanation.
** Linear fit disregards erroneous early temperature readings.

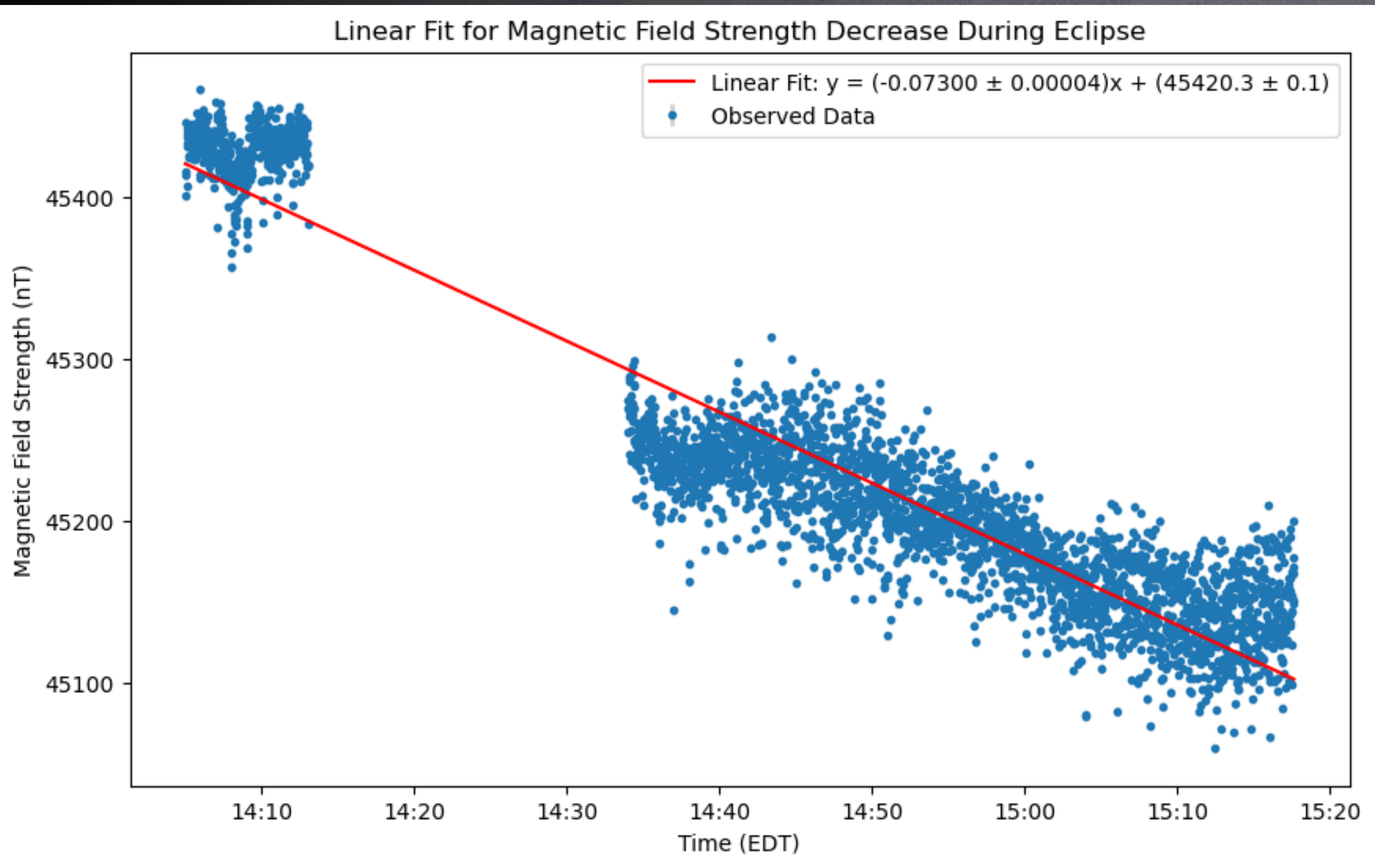
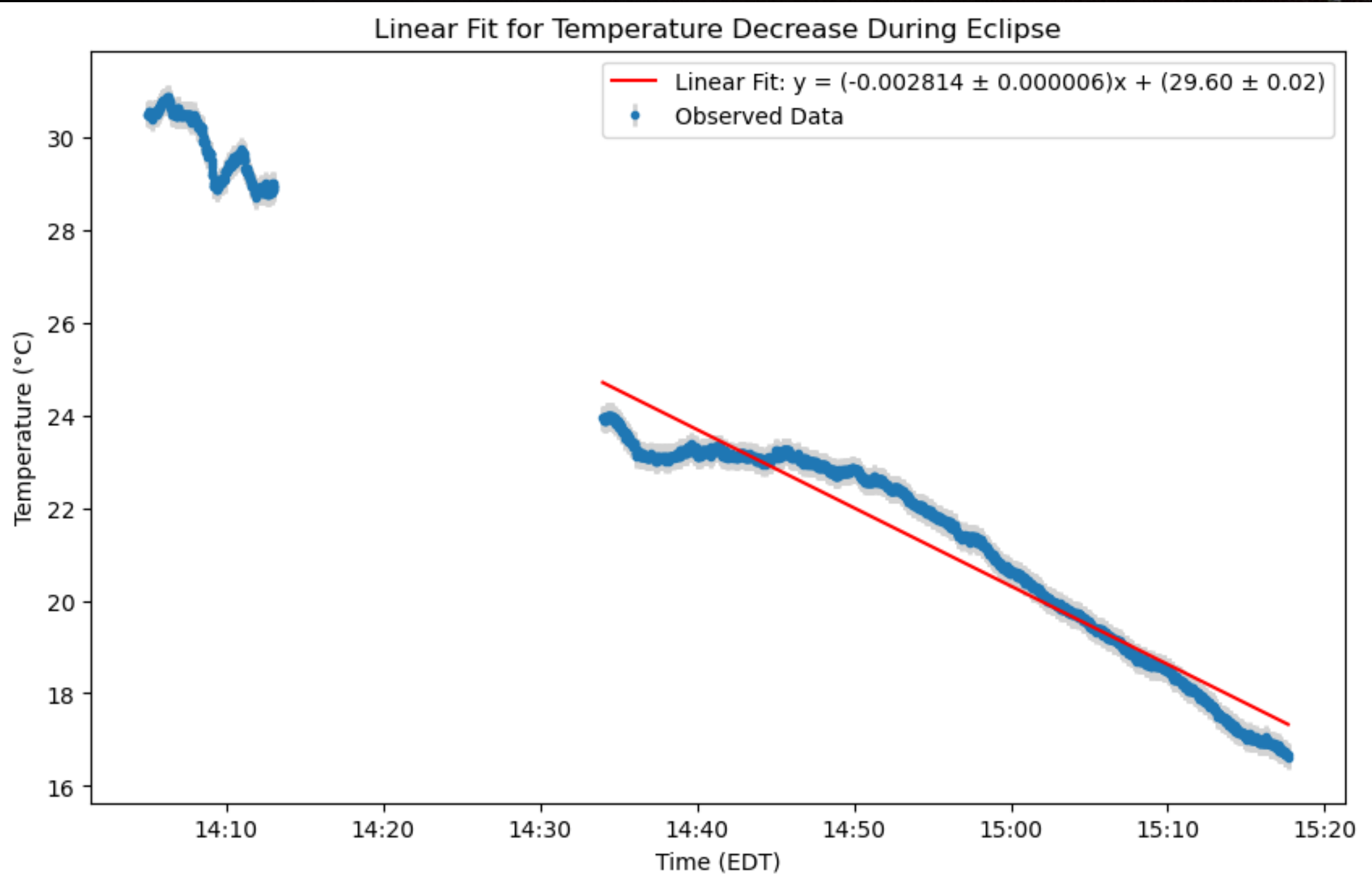


Table 1: Changes in Magnetic Field Strength and Temperature

Data Type	Peak-to-Peak	Rate of Change
Magnetic Field Strength (nT)	407 ± 4 nT	-0.07300 ± 0.00004 nT/s
Temperature	7.9 ± 0.4 °C*	-0.002814 ± 0.000006 °C/s**

Figure 1: Image of eclipse taken during totality

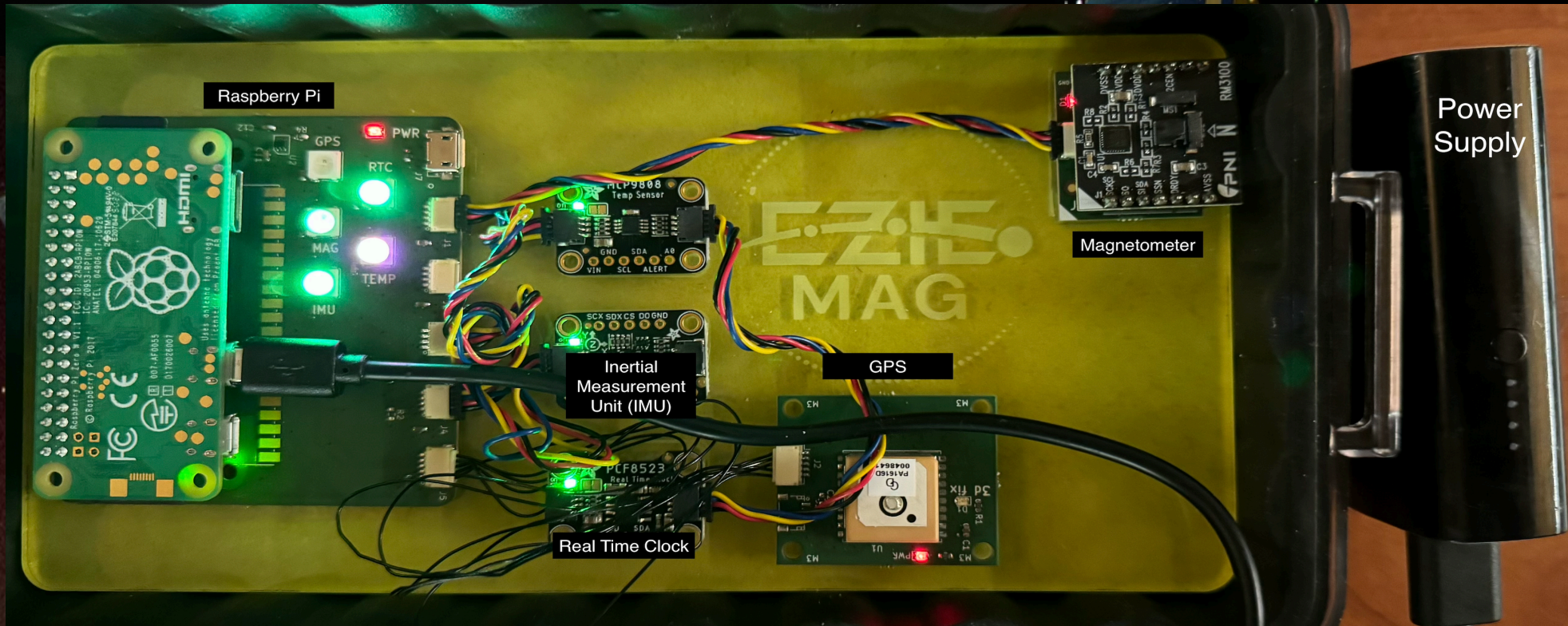


Figure 2: EZIE-Mag apparatus

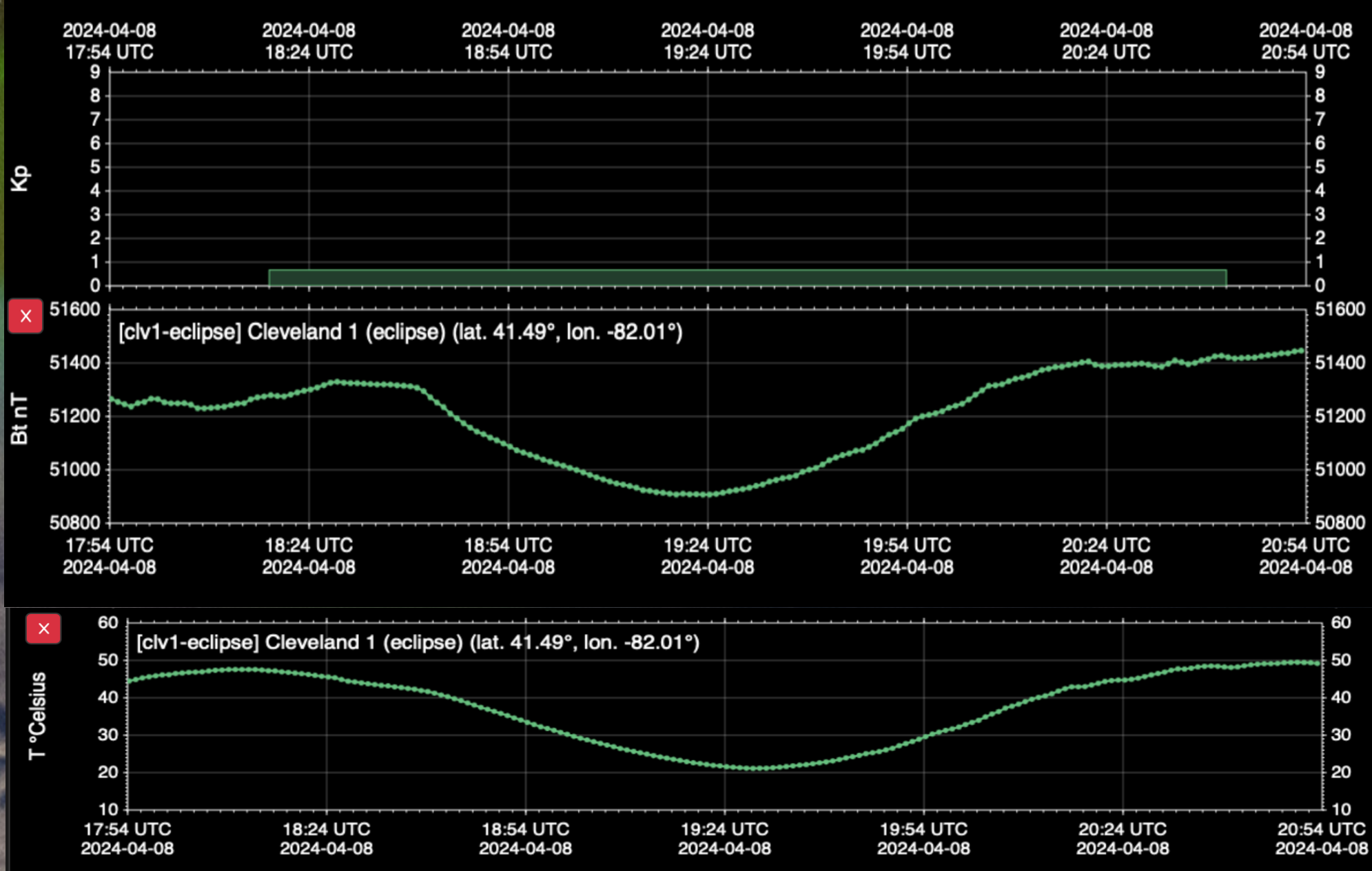


Figure 3: Cleveland EZIE-Mag Readouts Along the Path of Totality

Discussion

We conclude that during the eclipse, the temperature dropped 7.9 ± 0.4 °C, which makes sense given to the blockage of solar radiation that warms the earth. Interestingly, a decrease in magnetic field strength was also observed, generally lowering as the sun got more blocked.

Another EZIE-Mag kit was located 49 miles away in Cleveland and measured similar thermal and electromagnetic activity (see Figure 3). It seems to have abnormally high recorded temperatures, possibly from electronic heat dissipation. Our earlier readings were similar and have been disregarded in our calculations. Despite our local electromagnetic readings, the Kp-index, a measure of the magnitude of geomagnetic storms, was low. Values of Kp above 4 indicate active geomagnetic conditions, and a Kp value of 8 or 9 indicates an extreme geomagnetic storm.